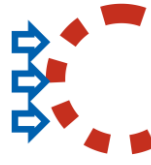
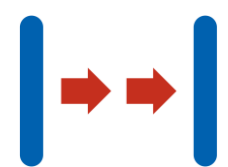
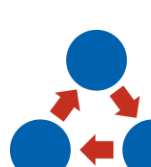
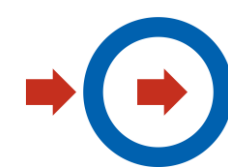
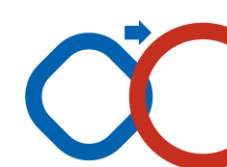
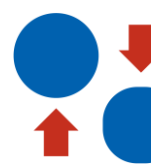
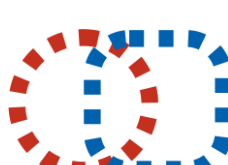
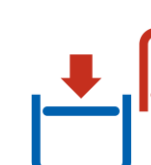
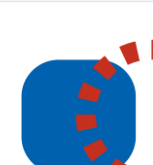


01  SEGMENTATION WHAT HAPPENS IF YOU DIVIDE AN OBJECT? 1. divide the object into independent pieces 2. make the object easy to decompose 3. reinforce the degree of fragmentation e.g., pizza in pieces, furniture from Ikea	02  SEPARATE WHAT PART CAN YOU SEPARATE? 1. separate the part of the object or the property/function itself that is in question 2. separate out only the necessary part or property of the object e.g., dietary supplements, perfume, contact lenses, franchise business, Google ads	03  LOCAL QUALITY HOW COULD ANY PART OF THE FACILITY BE CHANGED? 1. change the structure of the facility or environment from homogeneous to heterogeneous 2. let each part of the facility function in the best conditions for itself 3. let each part of the object perform a different and useful function e.g., a straw that can be bent, a perforation that makes it easier to tear a piece of paper, a date magnification in a watch, a magnetic tip in a screwdriver, a glove for touch screens	04  ASYMETRY WHAT HAPPENS IF YOU CHANGE AN OBJECT ASYMMETRICALLY? 1. change the shape of the object from symmetrical to asymmetrical 2. if the object is asymmetrical, increase the asymmetry e.g., Tripod, Boeing 737 engine air intake, Senz storm umbrella, asymmetrical earphone cable	05  CONNECTING IS IT POSSIBLE TO COMBINE OBJECTS SO THAT THEY PERFORM OPERATIONS SIMULTANEOUSLY? 1. bring identical or similar objects closer or together, assemble similar/identical objects together so that they work together 2. let there be a combined or parallel operation, join them together in time e.g., medical tour, all-in-one shampoo, hypermarket, charity concert, shopping Sunday	06  MULTIFUNCTIONALITY IS IT POSSIBLE FOR AN OBJECT TO PERFORM MULTIPLE FUNCTIONS? 1. make the object or structure such that it performs more functions, make other parts redundant e.g., Swiss Army pocketknife, smartphone, multifunctional fitness equipment, sofa that folds into a bed, two-way air conditioning
07  MATRYOSHKA - ONE INSIDE THE OTHER WHAT HAPPENS IF YOU PUT ONE IN THE OTHER? 1. put one object into the other or put one object into the other one by one 2. let one part pass through the free space in the other e.g., fireman's ladder, hamburger, ATM inside a bank, customized furniture	08  COUNTERWEIGHT IS THERE ANY COUNTERWEIGHT TO OFFSET THE WEIGHT OF THE OBJECT? 1. to compensate for the weight of the object combine it with others to cause ascension 2. to compensate for the weight of the object let it interact with the environment e.g., counterbalance for elevator cab, forklift, product placement as advertising	09  PREVIOUS COUNTERACTION HOW CAN YOU TAKE A STEP BACK TO TAKE TWO STEPS FORWARD? 1. make an action with both harmful and useful effects 2. make a prior stress on the object e.g., a vaccine, self-reporting to the police, a stand with free samples, a free trial period, a call for the return of products from a series in which a defect has been detected	10  PRIOR ACTION WHAT CAN BE DONE BEFOREHAND? 1. make the necessary object change before it is needed (partially or completely) 2. arrange objects so that they are ready for action right away e.g., an interim a return envelope, stickers on a phone that detect moisture, cut on a kitchen paper towel, a reservation,	11  ADVANCE COMPENSATION HOW CAN YOU PREPARE FOR AN EMERGENCY SITUATION? 1. prepare for an accident - compensate in advance for the object's contingency e.g. car airbag, spare wheel, fuse, insurance	12  LEVELING UP CAN THE CONDITIONS OF OPERATION BE CHANGED TO MAKE OBJECTS HORIZONTALLY LEVEL? 1. change the conditions to eliminate the need to raise or lower objects in the gravity field e.g., thicker shoe insole, movie credits, low-floor bus, children's menu in restaurant, height-adjustable chair
13  REVERSE CAN IT BE DONE IN REVERSE? 1. do the activity in reverse 2. make the moving parts stationary and the non-moving parts moving 3. turn an object or process "upside down,, e.g., library on wheels, homework, door-to-door sales, slow food	14  CURVES / ROUNDINGS WHAT HAPPENS IF YOU USE CURVILINEAR OBJECTS? 1. instead of rectilinear parts, use curvilinear ones 2. instead of rectilinear motion, use rotary motion 3. use rollers, circles, spirals, domes, circularity e.g., traffic circle, arches in architecture, rotary table, spiral stairs, revolving doors	15  DYNAMIZATION WHICH PARTS CAN MOVE AGAINST EACH OTHER? 1. let the characteristics of the object find its optimal working conditions 2. If the object is rigid or inflexible, make it movable or flexible 3. divide the object into parts that can move relative to each other e.g., a gimbal, a folding bicycle, extra seats in a car that are removed from the floor, a folding/rolling keyboard	16  PARTIALLY OR EXCESSIVELY WHAT HAPPENS IF THERE IS A LITTLE LESS OR MORE OF SOMETHING? 1. if it is difficult to achieve 100% of an object by a certain way then use a little more or less of that way e.g. buffet restaurant, post-it notes, special edition (limited), Beta version (underdeveloped)	17  CHANGE OF DIMENSION WHAT WILL BE IN THE SPACE OF OTHER DIMENSIONS? 1. use an additional dimension in space 2. make a multilayer arrangement of objects instead of a single-layer arrangement 3. tilt the objects or reorient them spatially e.g., multi-level parking lot, QR code, 3D movie, 3D booklet, Wall-mounted TV	18  VIBRATIONS WHAT HAPPENS IF YOUR SUBJECT OBJECTS TO VIBRATION? 1. cause the object to oscillate or vibrate 2. increase the frequency of the vibration, such as to ultrasound 3. use the object's own resonant frequency e.g. phone vibration, ultrasonic cleaning, ultrasound, ultrasonic humidifier
19  PERIODIC ACTION WHAT HAPPENS IF PERIODIC ACTION IS USED? 1. instead of continuous action, use periodic or pulsating action 2. if the action is already periodic, change the amplitude or frequency 3. use pauses between pulses to perform another action e.g., installment purchase, traffic regulation by traffic lights, automatic air freshener, lane shifted by the dispatcher to the needed direction	20  CONTINUOUS USEFUL ACTION WHAT WILL BE DONE IF THE OPERATION IS CONTINUOUS? 1. conduct continuous operation. Let all parts work at maximum load 2. eliminate idle movements or intermittent operations 3. switch from reciprocating to rotating motion e.g. 24-hour store, traffic circle, greenhouse, indoor ski slope, indoor stadium	21  ACCELERATE - JUMP WHAT WILL HAPPEN TO OBJECTS AT HIGH SPEED? 1. run a process or certain steps at higher speeds to avoid what is harmful, destructive or risky e.g., non-stop highway toll collection, drive-thru, UHT (seconds) pasteurization, high-speed freezing, ultra-fast cutting, RFID chips	22  TURN HARM INTO BENEFIT IS IT POSSIBLE TO USE HARMFUL FACTORS TO ACHIEVE POSITIVE EFFECTS? 1. use harmful factors to achieve beneficial effects 2. eliminate the original harmful effect by combining it with another harmful effect 3. amplify the harmful factor to such an extent that it is no longer harmful e.g. recovering waste materials, hiring hackers as security consultants, Botox	23  FEEDBACK WHAT HAPPENS IF YOU ADD FEEDBACK? 1. introduce feedback 2. if the feedback already exists then change its strength or impact e.g., evaluation survey, Facebook likes, article comments, checking questions on participants before including them in a scoring test, parking lot referral system, floats	24  INTERMEDIATE WHAT INTERMEDIARY OBJECT CAN TEMPORARILY PERFORM THE ACTION? 1. use an intermediary or intermediate process 2. temporarily connect one object to another that can be easily removed e.g., shoe spoon, medical capsules, chauffeur, real estate broker, absorbent for oil stain removal
25  SELF-SERVICE HOW CAN YOU AUTOMATE? 1. let the facility operate itself by performing auxiliary operations 2. use waste or wasted resources, substances or energy e.g., urinal sensor, self-assembling furniture, automatic faucet, self-service stand, self-driving robot vacuum cleaner, electronic banking	26  COPYING ISN'T IT ENOUGH TO USE COPIES? 1. instead of inaccessible, expensive or fragile objects, use simpler and cheaper copies 2. turn the object into an optical copy e.g., artificial flowers, mannequins at an exhibition, a scarecrow, a simulator, a video conference	27  PERISHABLE / NOT DURABLE BUT CHEAP WHAT CAN BE REPLACED WITH DISPOSABLES? 1. replace an expensive object with many inexpensive objects by reducing the quality locally e.g., disposable dishes or lenses, diapers, syringes	28  REPLACING MECHANICAL INTERACTION WHAT OTHER SENSORY APPROACH CAN REPLACE THE MECHANICAL ONE? 1. replace mechanical means with sensory ones (optical, acoustic, taste, smell) 2. use fields - electric, mechanical or electromechanical to react with the object e.g., books for the blind, an acoustic signal at traffic lights, a warning smell added to gas, an ultrasonic animal repellent	29  PNEUMO AND HYDRO STRUCTURES CAN YOU USE GAS OR LIQUID INSTEAD OF SOLID PARTS? 1. use inflatable or liquid-filled parts e.g., inflatable tent dome, waterbed, bubble wrap, inflatable collar/pillow for a long trip	30  FLEXIBLE COATINGS AND THIN FILMS CAN YOU INSULATE/PROTECT AN OBJECT USING THIN FILMS? 1. use flexible sheaths and thin films instead of spatial structures 2. isolate the object from harmful external influences using flexible covers or thin films e.g., air curtain, film covering the screen of a smartphone, UV filter cream, window tinting with a special film
31  POROSITY WHAT HAPPENS IF YOU USE POROUS MATERIALS? 1. make the object porous or add porous elements 2. if the object is already porous, use the space in these pores for a useful purpose e.g. insect netting, water filter, silicate gel (hygroscopic), foam soap	32  CHANGING OPTICAL PROPERTIES IS IT POSSIBLE TO CHANGE THE COLOR OR TRANSPARENCY? 1. change the color of an object or its external environment 2. change the transparency of an object or its external environment e.g. litmus paper, stickers that change color when exposed to temperature, transparent wrapping film, photochromic glasses, red stamp on Tefal pans, transparent button in an application	33  UNIFORMITY WHAT HAPPENS IF THE OBJECTS ARE SIMILAR OR THE SAME? 1. let the objects react with a given object of the same material or with identical properties e.g., a pair of wedding rings, a camouflage uniform, a street of cobblers, plates of bran (an edible bite), skin-colored bandages	34  REJECTION AND REGENERATION IS IT POSSIBLE TO DISCARD OR REGENERATE AN OBJECT? 1. let the parts of the system that have already fulfilled their function drop off or modify them during operation 2. conversely, regenerate consumables during operation e.g., multi-member rockets, absorbable surgical threads, refillable cartridges, second-hand stores, rechargeable batteries	35  PARAMETER CHANGE IS IT POSSIBLE TO CHANGE THE PARAMETERS OF AN OBJECT? 1. change the physical state of the object (solid, gas, liquid) 2. change the consistency or concentration 3. change the level of elasticity e.g. popcorn, liquid soap, glue gun, liquid oxygen, wet wipes, LPG gas	36  PHASE TRANSITION CAN YOU USE THE PHASE TRANSITION? 1. use the phenomena accompanying the phase transition (volume change, deformation, heat transfer) e.g., air conditioning, sandblasting with dry ice, perfume, heating pipe, sprinkling salt on ice, steam engine
37  THERMAL EXPANSION WHICH SPECIFIC PART DO YOU WANT TO EXPAND? 1. use the thermal expansion of materials 2. if you already use it, use the size of the multiplicity of materials with different properties e.g. hot air balloon, mercury thermometer, bimetal, expansion spaces	38  ENHANCED INTERACTION WHAT HAPPENS IF WE ADD SOMETHING REINFORCING? 1. instead of air, use oxygen 2. instead of oxygen, use ozone e.g. catalyst, ozone sterilization, fermentation, cheerleaders, bonus or premium	39  NEUTRAL ENVIRONMENT WHAT HAPPENS IF INERT PARTICLES ARE ADDED? 1. replace the environment with a neutral one 2. add something neutral or neutralizing additives to the object e.g., aseptic room, nitrogen environment packaging, fire extinguisher, welcome page on Google, vacation	40  COMPOSITE MATERIALS CAN NEW PROPERTIES BE OBTAINED BY JOINING OBJECTS? 1. change a homogeneous material into a composite (multi composite) structure e.g., reinforced concrete, orchestra, bouquet of flowers, plywood, car window	40 TRIZ PRINCIPLES Material prepared and reproduced with permission of ODITK (based on Polish version) available at https://www.triz.oditk.pl/40-zasad	